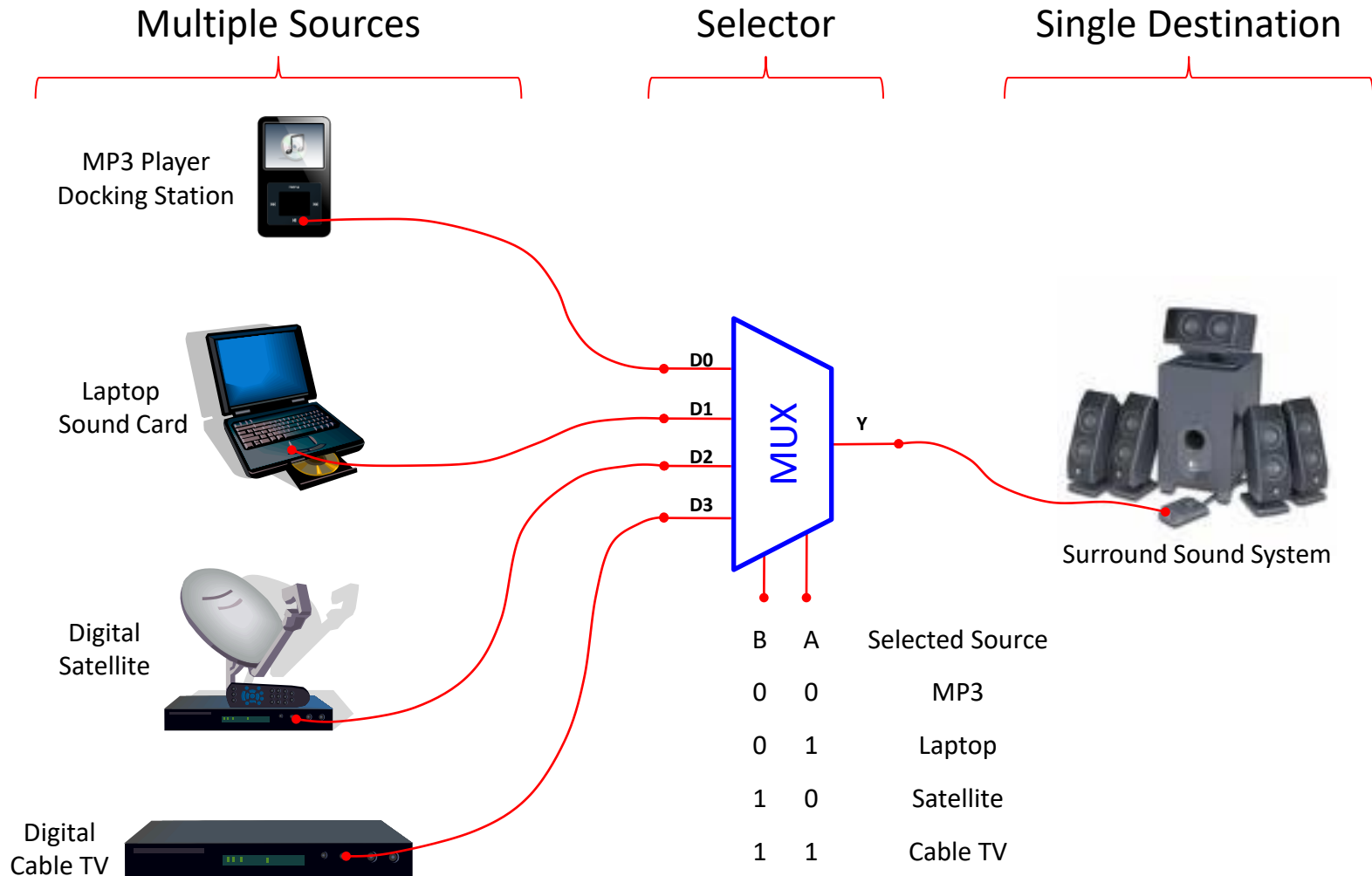


Multiplexer / Demultiplexer

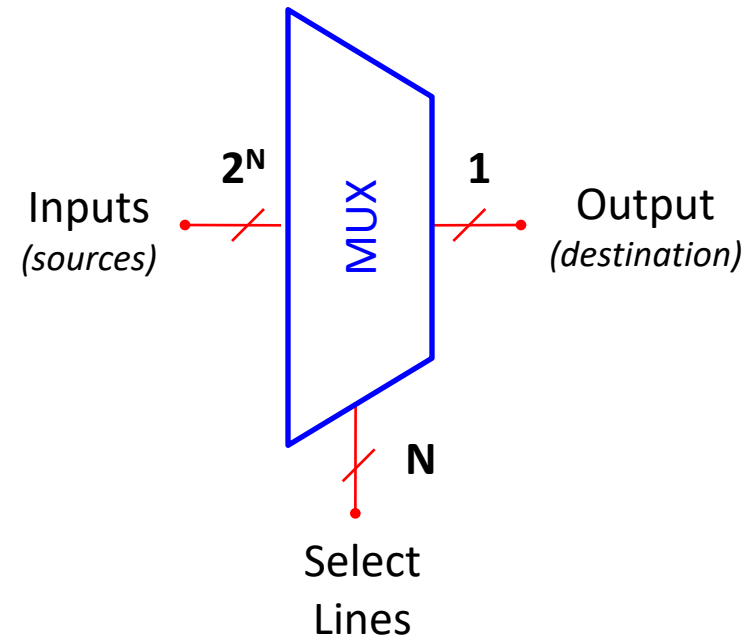
Typical Application of a MUX



What is a Multiplexer (MUX)?

- A MUX is a digital switch that has multiple inputs (sources) and a single output (destination). (Data selector)
- The select lines determine which input is connected to the output.
- MUX Types
 - 2-to-1 (1 select line)
 - 4-to-1 (2 select lines)
 - 8-to-1 (3 select lines)
 - 16-to-1 (4 select lines)

Multiplexer
Block Diagram

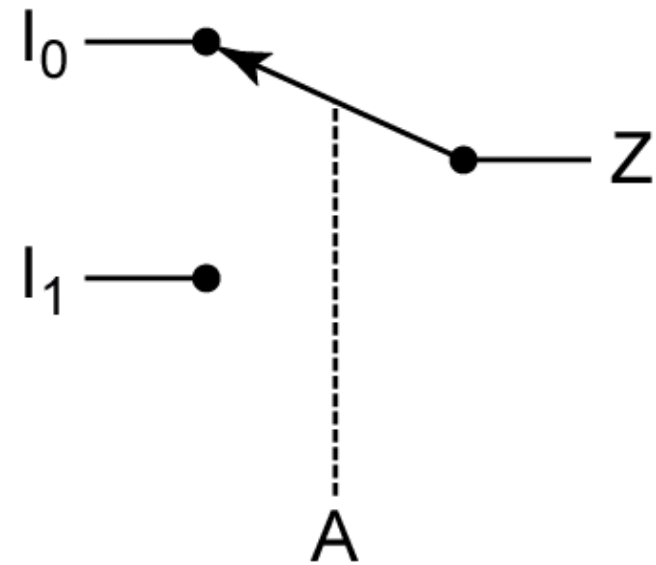
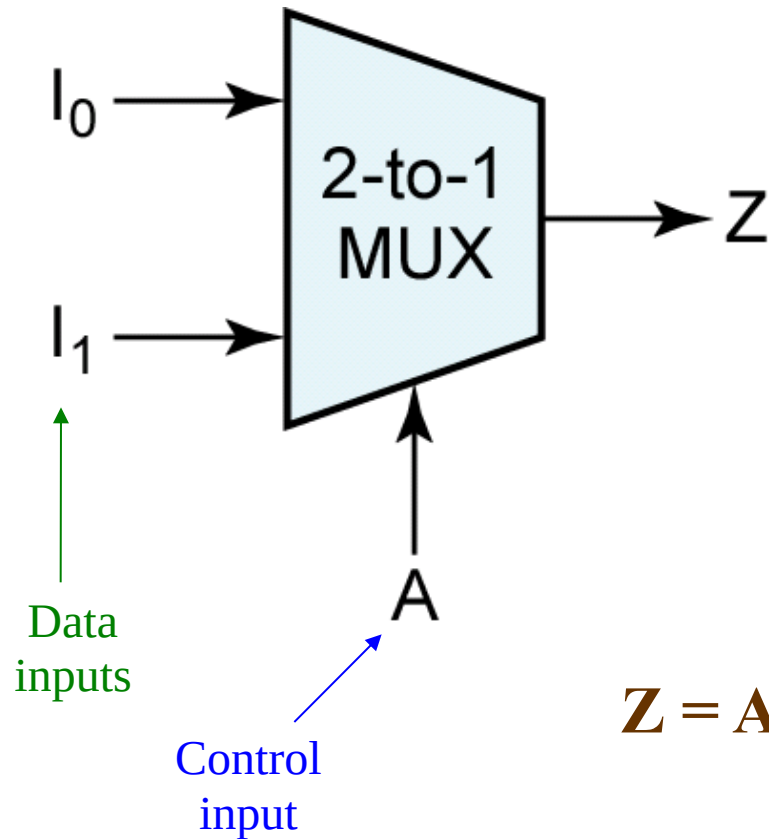


MULTIPLEXERS (Data Selectors)

A multiplexers (MUX) is a device that allows digital information from several sources to be routed onto a single line for transmission over that line to a common destination.

The basic multiplexers has several data input lines and a single output line. It also has data- select inputs, which permit digital data on any one of the inputs to be switched to the output line.

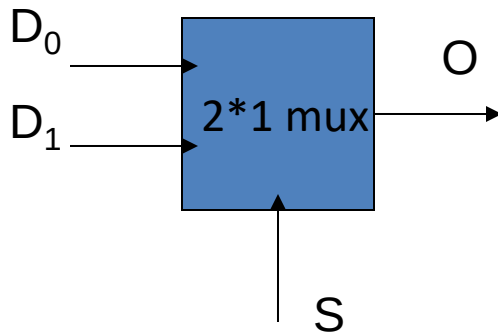
Multiplexers



$$Z = A'.I_0 + A.I_1$$

Design of a 2*1 Mux

- 2*1 mux Block Diagram
- Truth Table



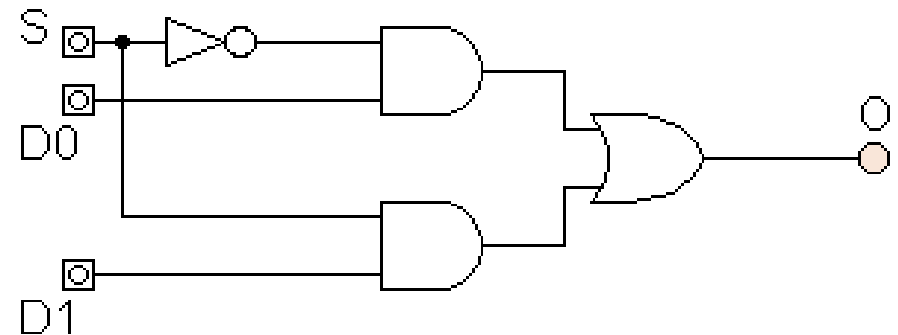
S	D ₁	D ₀	O
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Design of 2*1 Mux Continued

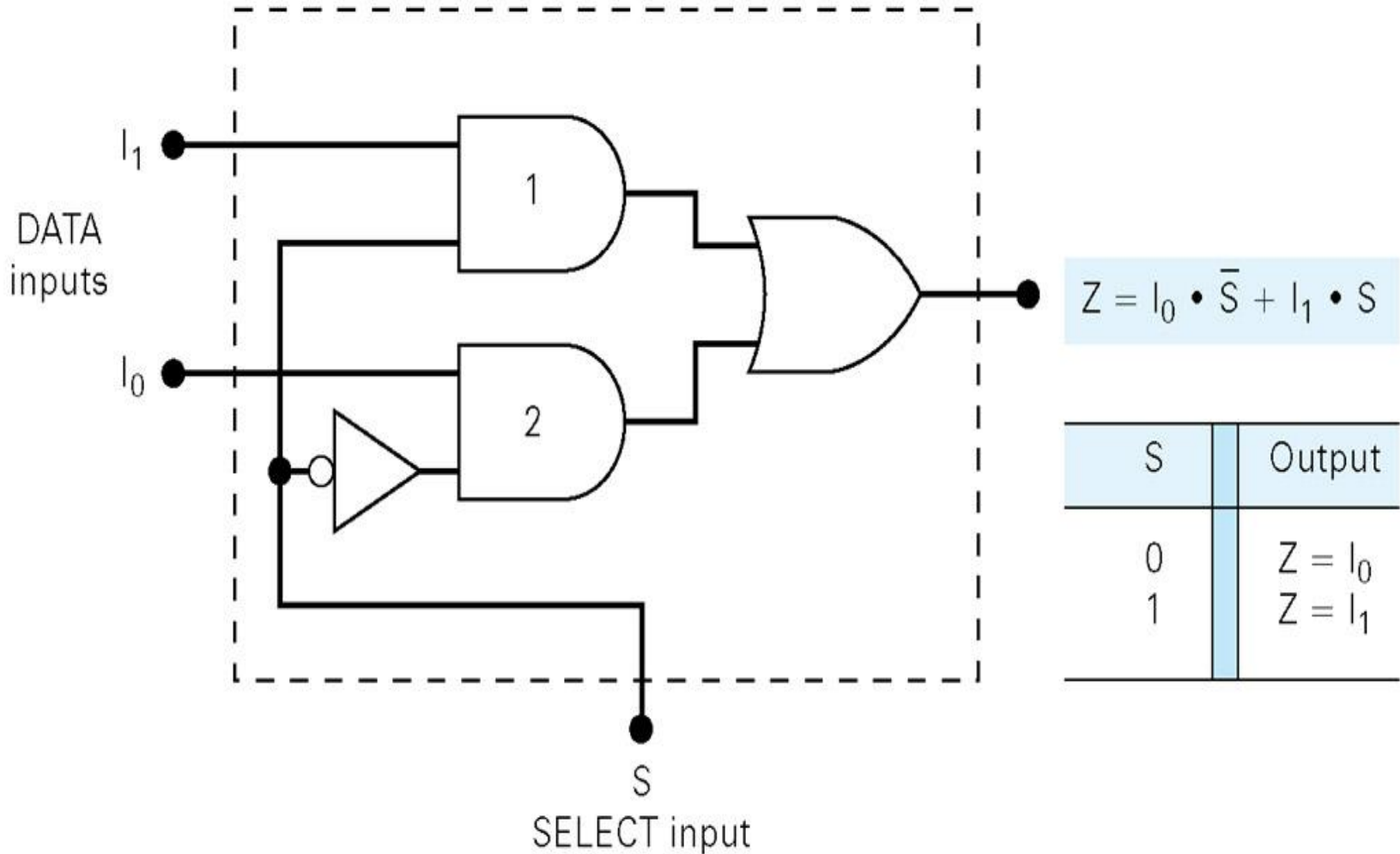
- K-Map

	$S'D_0$	01	11	10
D0				
0		1		
1		1	1	1

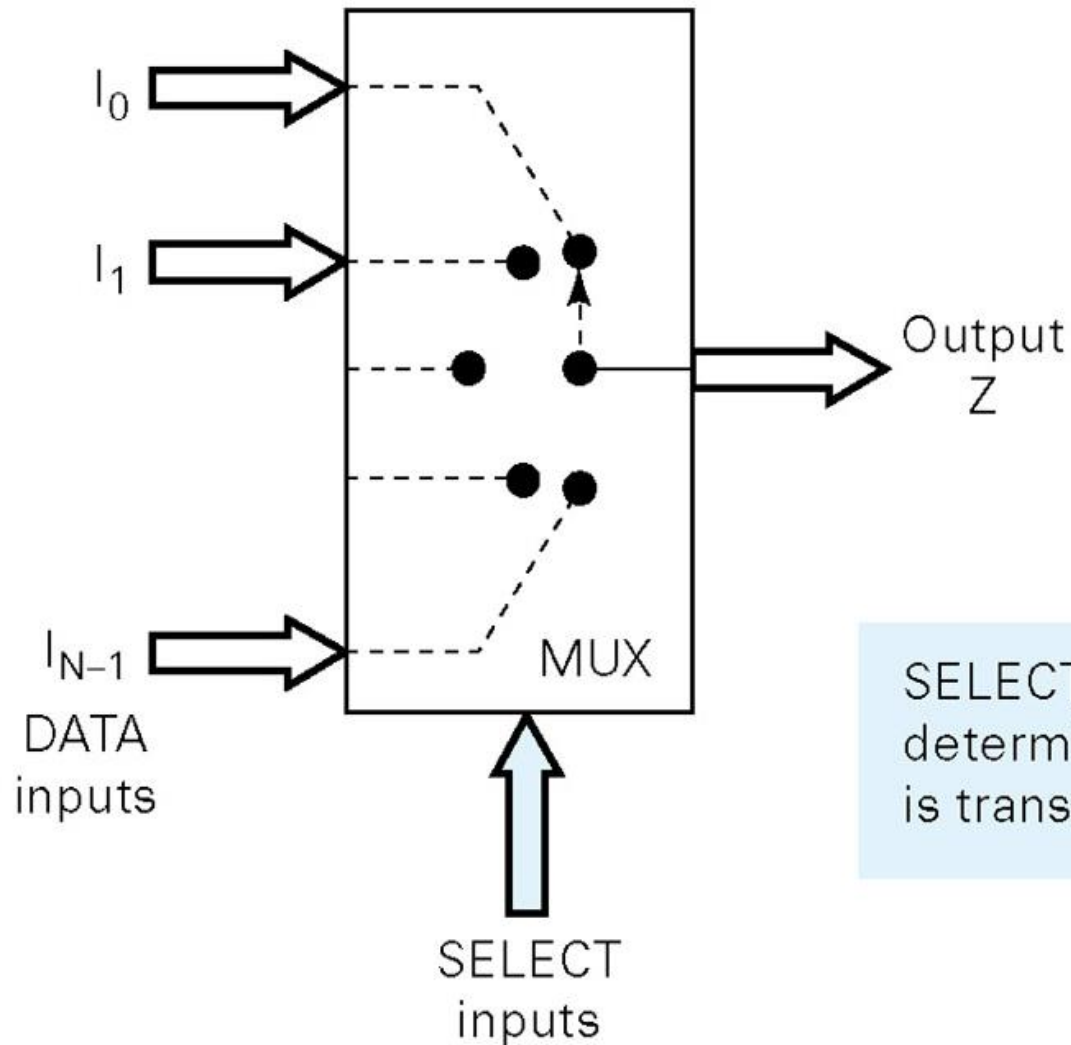
$$O = S'D_0 + SD_1$$



Two-input multiplexer

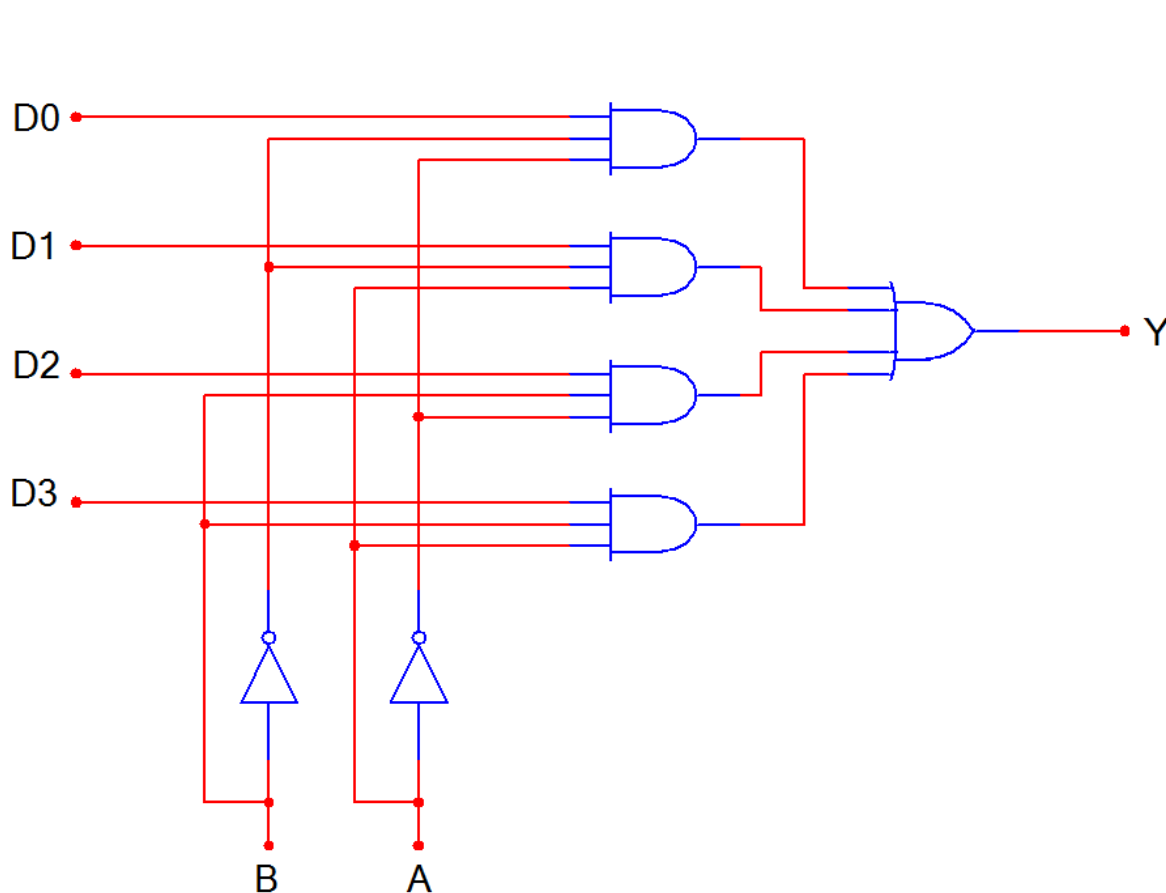


Functional diagram of MUX

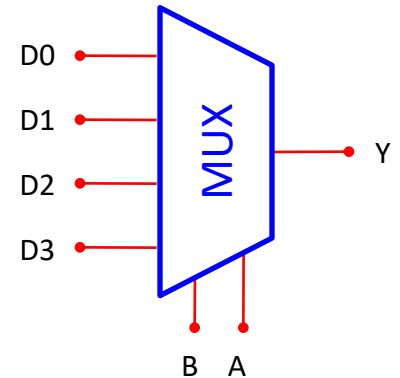


SELECT input code determines which input is transmitted to output Z.

4-to-1 Multiplexer (MUX)



$$Y = A'B'D0 + A'BD1 + AB'D2 + ABD3$$



A	B	Y
0	0	D0
0	1	D1
1	0	D2
1	1	D3

Uses of Multiplexers

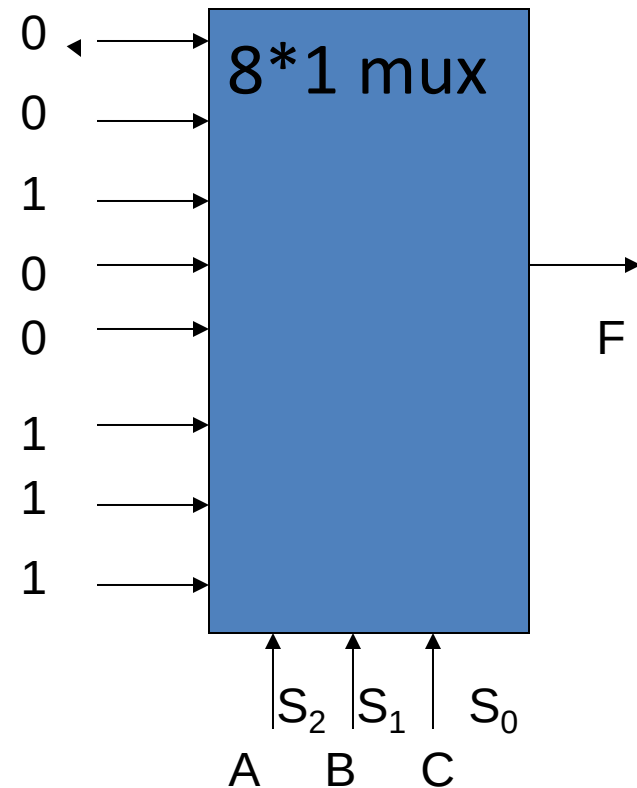
- Used in data communications for several computers to communicate over 1 line
- Used in radio to select one channel from many
- Used to route data within a computer
- Used for Logic function generation

Logic gate designing using MUX

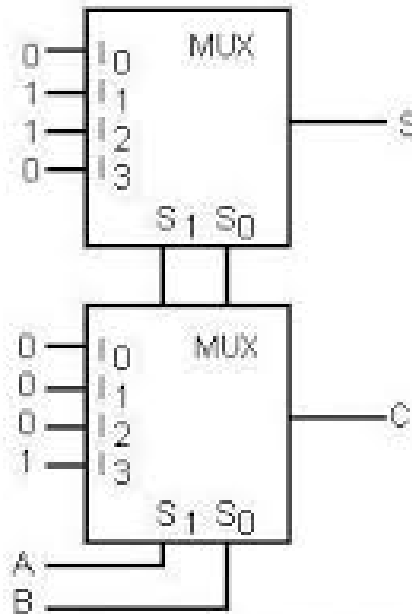
Logic function generation

Implement the Given truth table using a suitable MUX

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

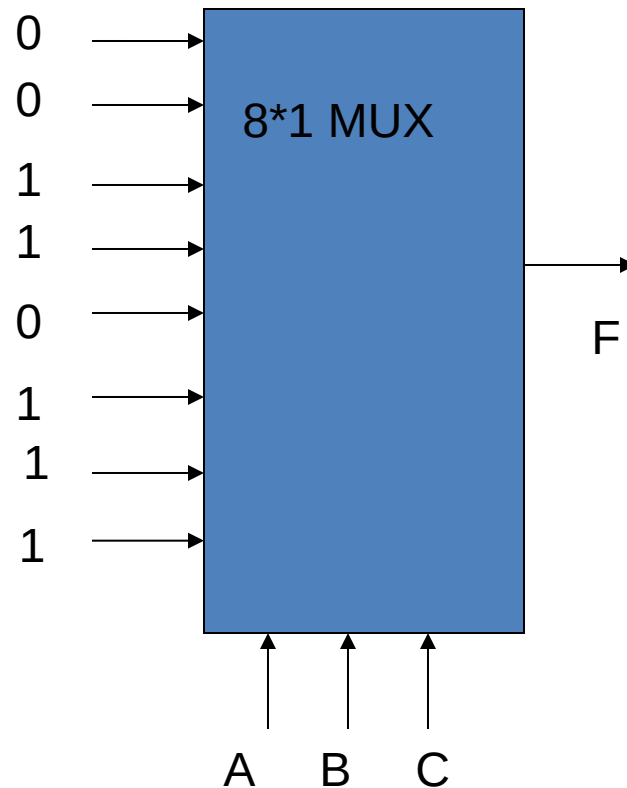


Implement the Half-adder circuit by using suitable multiplexer.



Using an 8-to-1 multiplexer, design a logic circuit to realize the following Boolean function

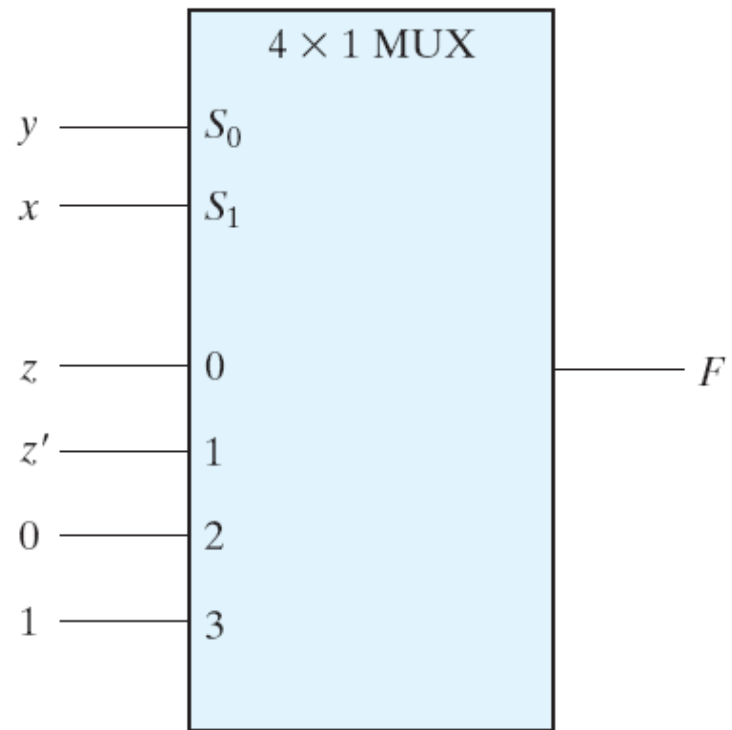
$$F(A,B,C) = \sum m(2, 3, 5, 6, 7)$$



$F(x,y,z) = \Sigma m(1, 2, 6, 7)$ using 4×1 mux

x	y	z	F	
0	0	0	0	$F = z$
0	0	1	1	
0	1	0	1	$F = z'$
0	1	1	0	
1	0	0	0	$F = 0$
1	0	1	0	
1	1	0	1	$F = 1$
1	1	1	1	

(a) Truth table

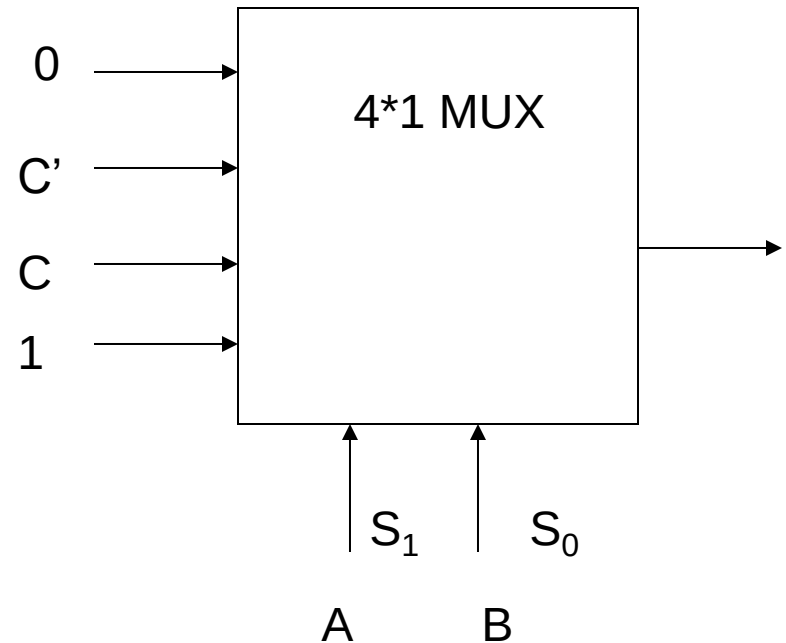


(b) Multiplexer implementation

Function Generation using Multiplexers

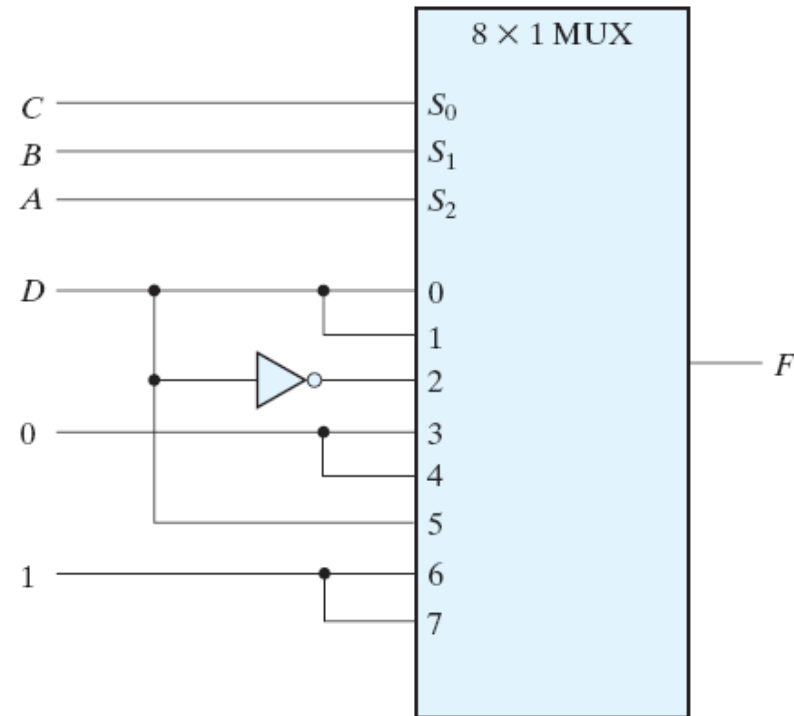
- Given the following truth table use a 4*1 Mux to generate F

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1



Example: $F(A,B,C,D) = \sum m(1,3,4,11,12-15)$ using 8×1 mux

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>F</i>	
0	0	0	0	0	$F = D$
0	0	0	1	1	
0	0	1	0	0	$F = D$
0	0	1	1	1	
0	1	0	0	1	$F = D'$
0	1	0	1	0	
0	1	1	0	0	$F = 0$
0	1	1	1	0	
1	0	0	0	0	$F = 0$
1	0	0	1	0	
1	0	1	0	0	$F = D$
1	0	1	1	1	
1	1	0	0	1	$F = 1$
1	1	0	1	1	
1	1	1	0	1	$F = 1$
1	1	1	1	1	



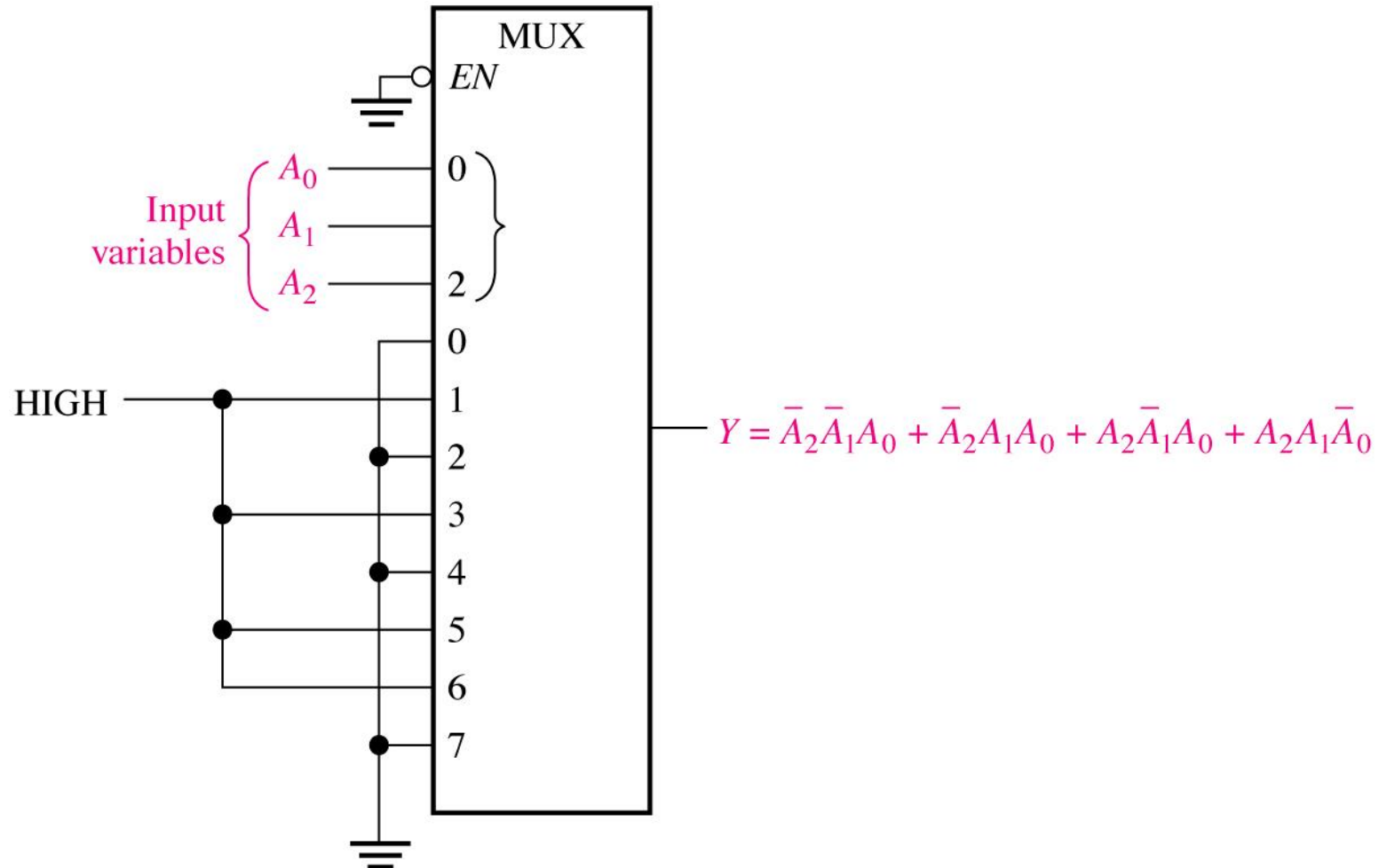
Example: $F(A,B,C,D) = \Sigma m(1,3,4,11,12-15)$ using 4*1 mux

LOGIC FUNCTION GENERATION USING MUX

Exercise 1:
Implement the logic circuit function specified in the table given below by using 74LS151 8-input data selector/multiplexer.

Input			Output	
A2	A1	A0	Y	
0	0	0	0	0
0	0	1	1	1
0	1	0	0	2
0	1	1	1	3
1	0	0	0	4
1	0	1	1	5
1	1	0	1	6
1	1	1	0	7

Solution :

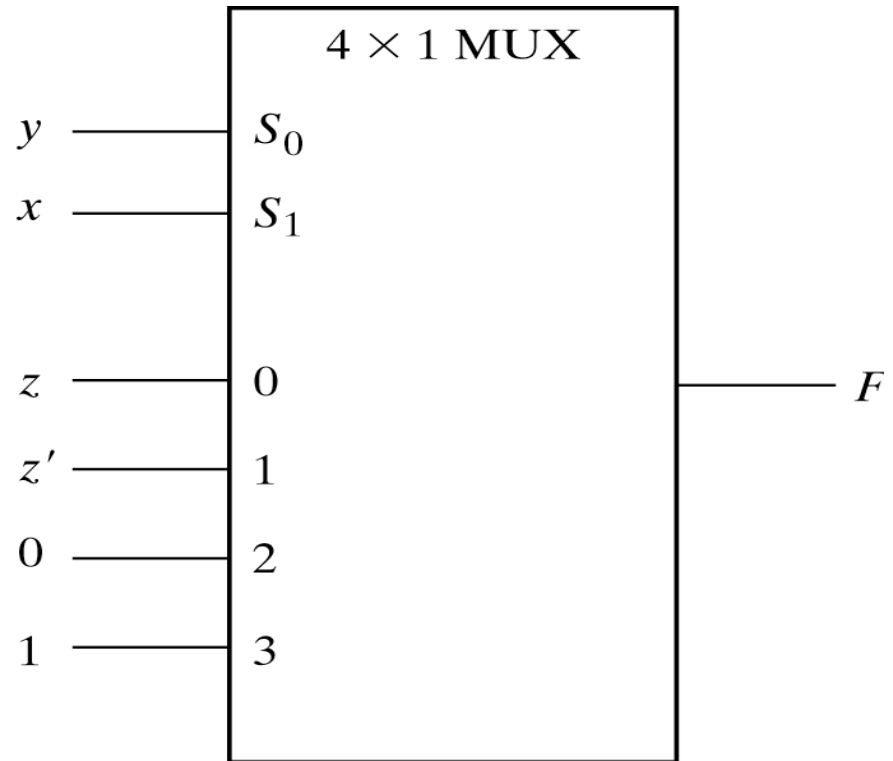


LOGIC FUNCTION GENERATION USING 4 * 1 MUX

Implement the Boolean function $F = x'y'z + x'yz' + xyz' + xyz$ using a suitable MUX

x	y	z	F		
0	0	0	0		
0	0	1	1	$F = z$	0
0	1	0	1		
0	1	1	0	$F = z'$	1
1	0	0	0		
1	0	1	0	$F = 0$	2
1	1	0	1		
1	1	1	1	$F = 1$	3

(a) Truth table



(b) Multiplexer implementation

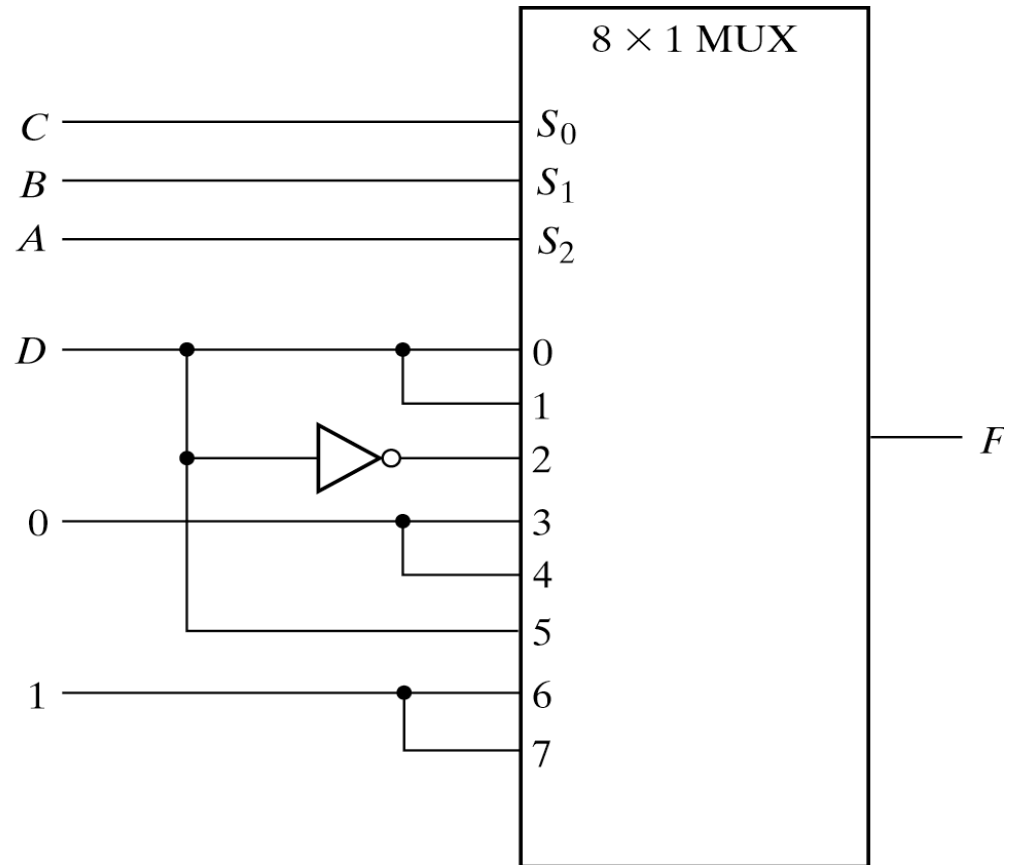
LOGIC FUNCTION GENERATION USING 8*1 MUX

Implement the Boolean function

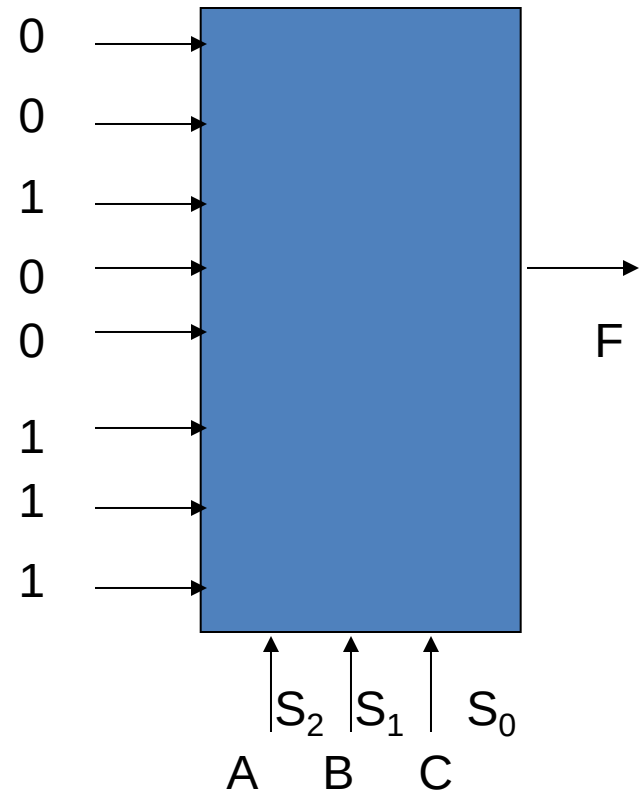
$$F = A'B'C'D + A'B'CD + A'BC'D' + AB'CD + ABC'D' + ABC'D + ABCD' + ABCD$$

using a suitable MUX and 8*1 Mux.

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>F</i>		
0	0	0	0	0		
0	0	0	1	1	$F = D$	0
0	0	1	0	0		
0	0	1	1	1	$F = D$	1
0	1	0	0	1		
0	1	0	1	0	$F = D'$	2
0	1	1	0	0		
0	1	1	1	0	$F = 0$	3
1	0	0	0	0		
1	0	0	1	0	$F = 0$	4
1	0	1	0	0		
1	0	1	1	1	$F = D$	5
1	1	0	0	1		
1	1	0	1	1	$F = 1$	6
1	1	1	0	1		
1	1	1	1	1	$F = 1$	7

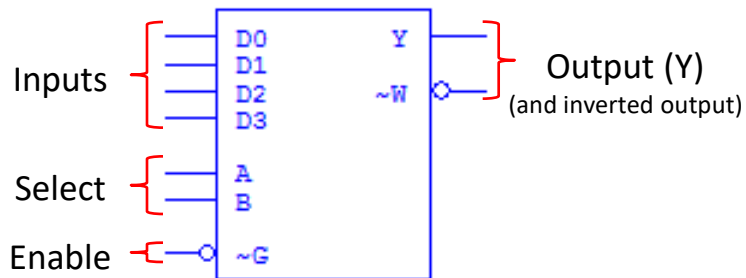


Find out the Boolean expression of given block diagram

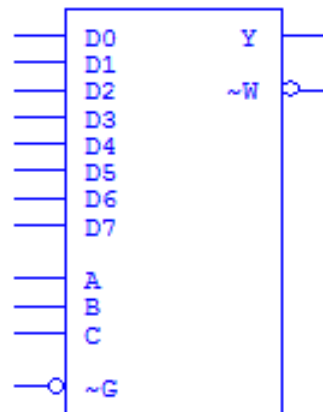


Medium Scale Integration MUX

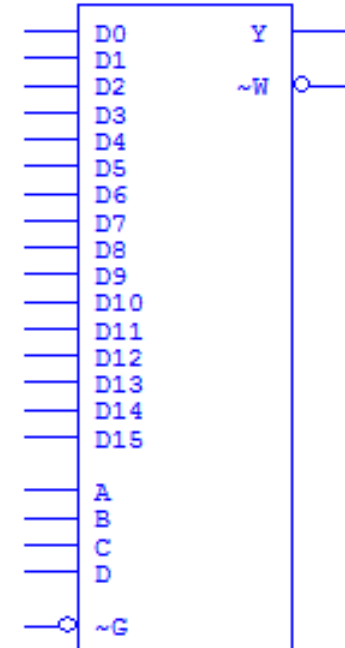
4-to-1 MUX



8-to-1 MUX

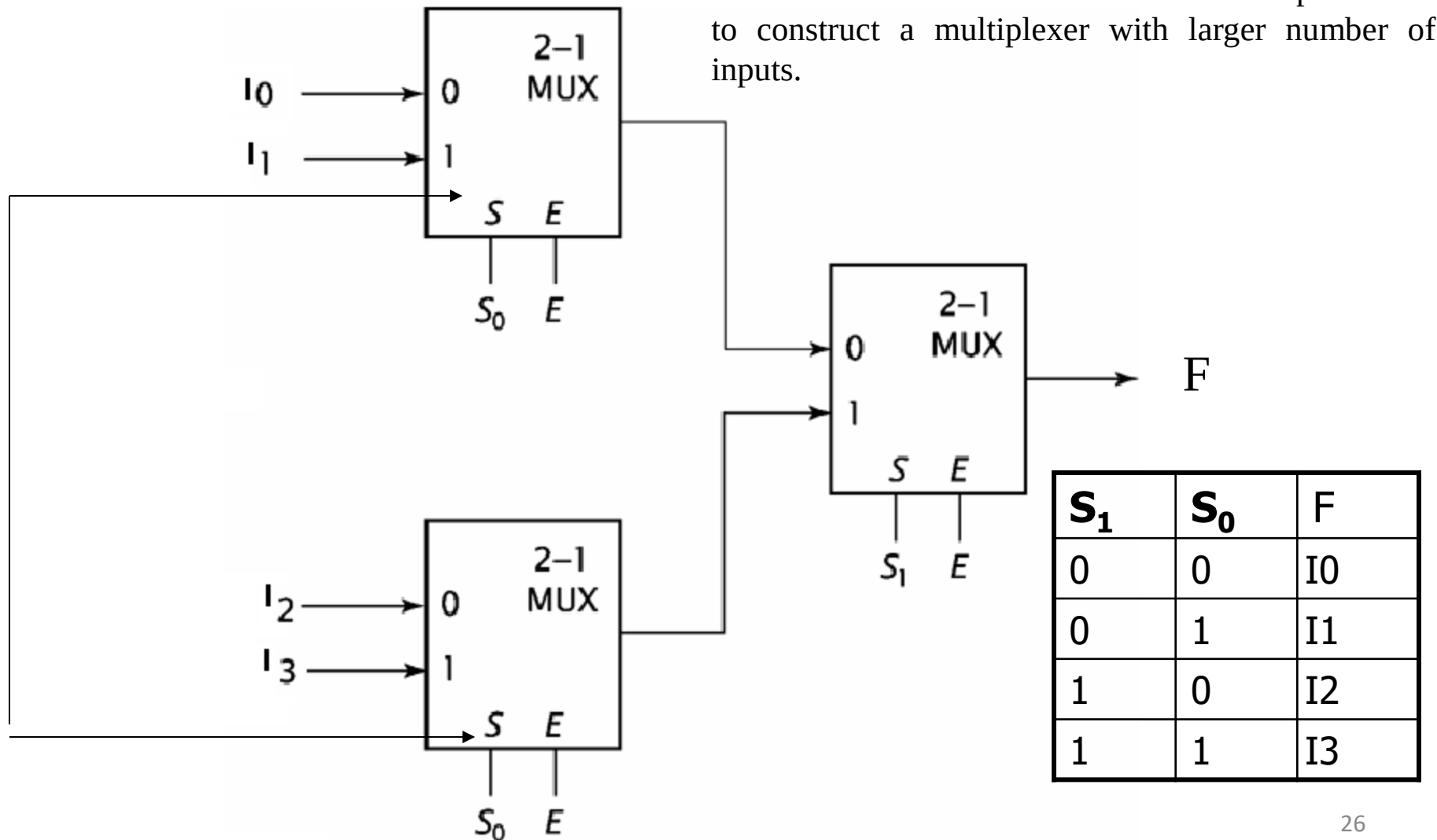


16-to-1 MUX



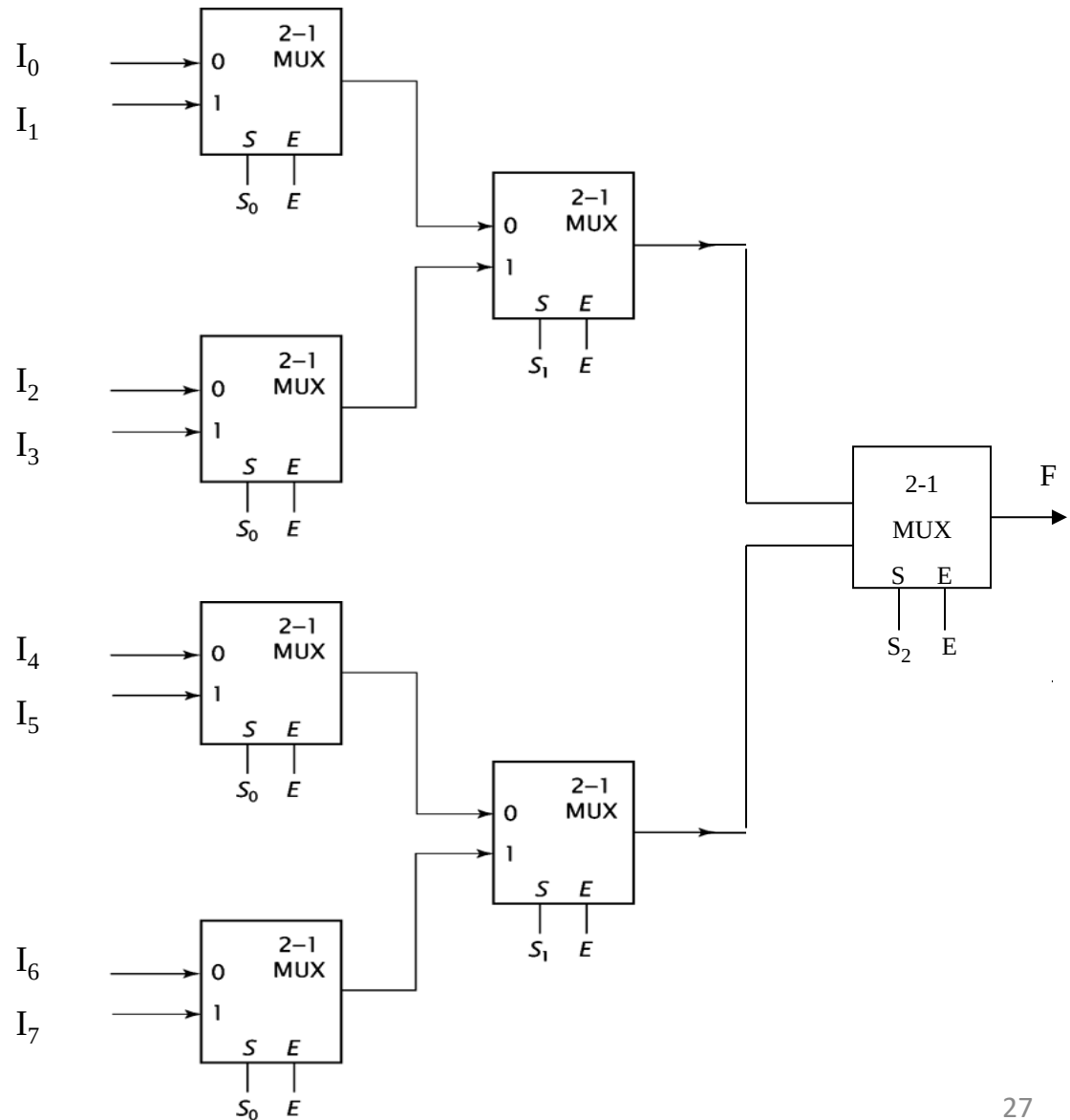
Cascading multiplexers/Mux Tree

Using three 2-1 MUX to make one 4-1 MUX

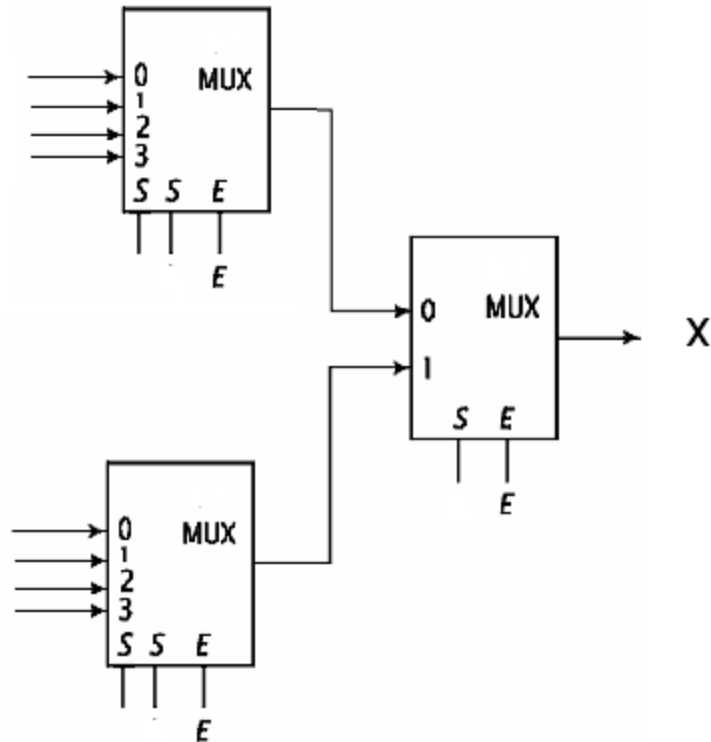


Construct an 8-to-1 multiplexer using 2-to-1 multiplexers.

S_2	S_1	S_0	F
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7



Example: Construct 8-to-1 multiplexer using one 2-to-1 multiplexer and two 4-to-1 multiplexers



S_2	S_1	S_0	F
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

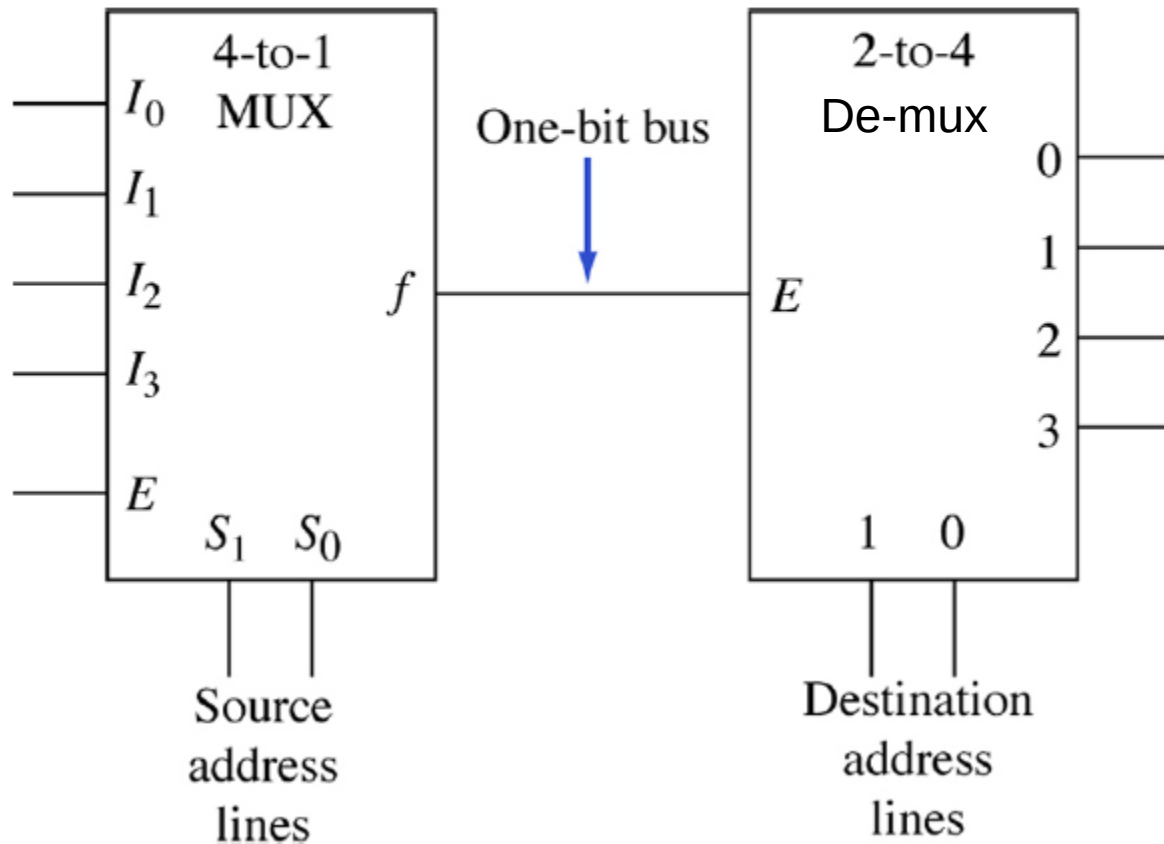
Example: Construct 8-to-1 multiplexer using only two 4-to-1 multiplexers and basic gates.

DEMULTIPLEXERS

A DEMULTIPLEXER (DEMUX) basically reverses the multiplexing function. It takes data from one line and distributes them to a given number of output lines. For this reason, the demultiplexers is also known as a data distributor.

A demultiplexer (DEMUX) performs the reverse operation of MUX. It takes a single input and distributes it over several outputs.

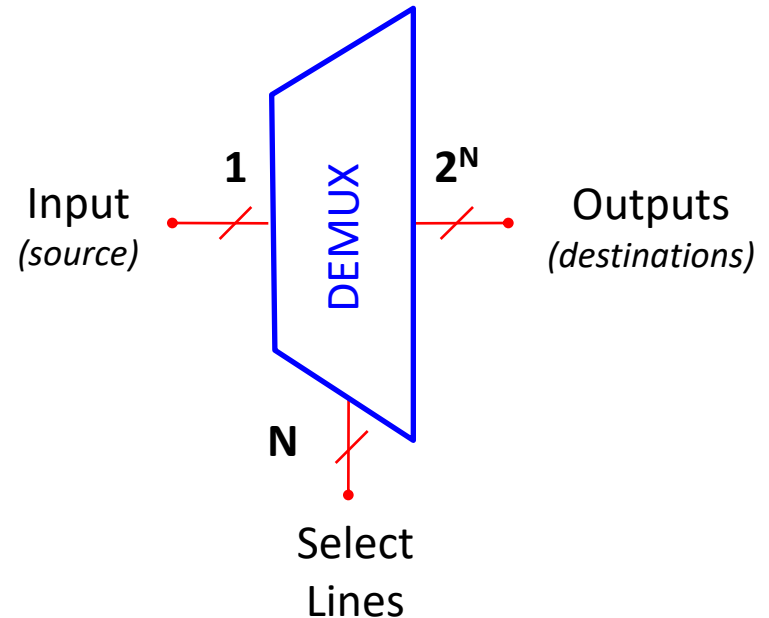
A multiplexer/demultiplexer arrangement for information transmission.



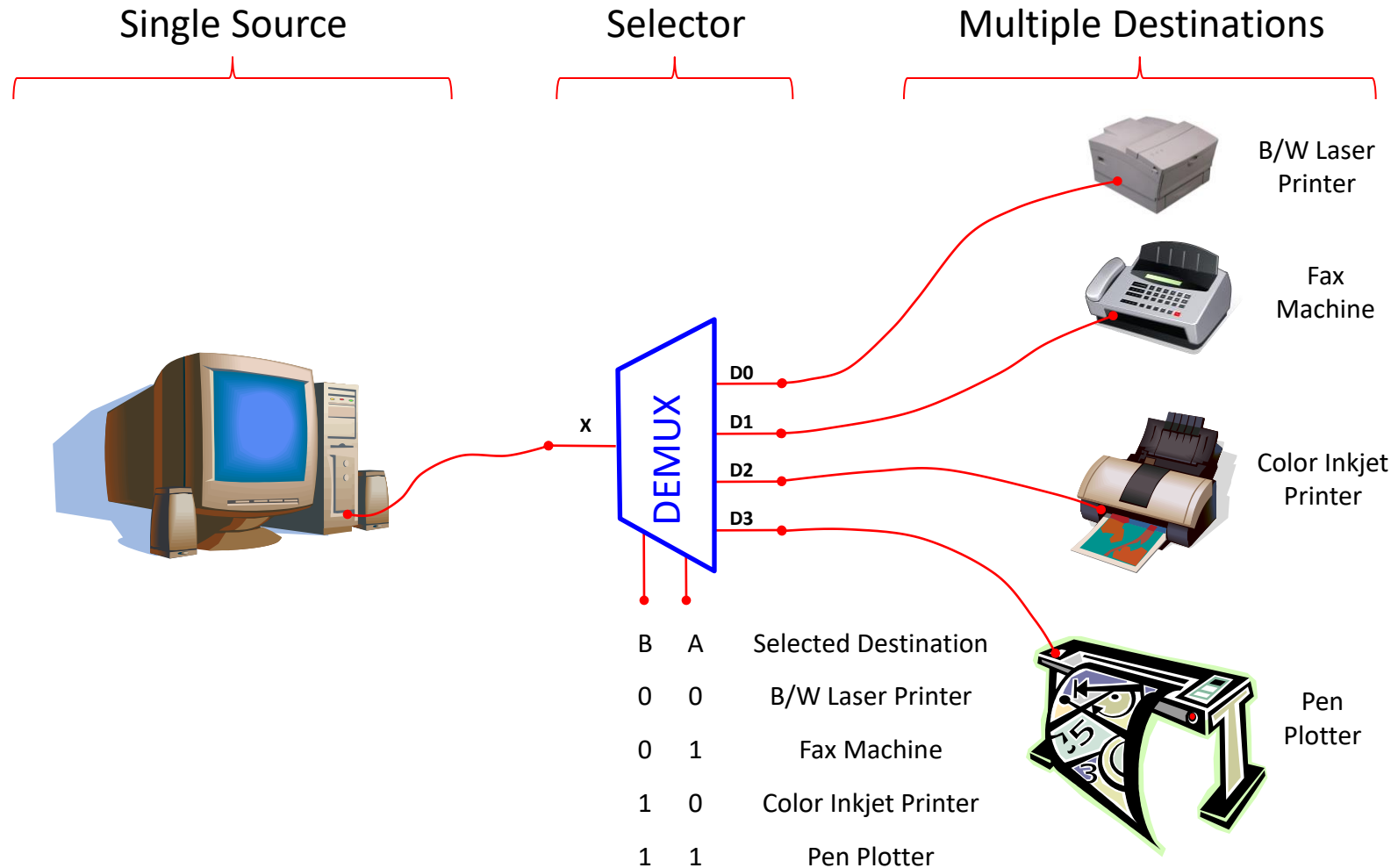
What is a Demultiplexer (DEMUX)?

- A DEMUX is a digital switch with a single input (source) and a multiple outputs (destinations).
- The select lines determine which output the input is connected to.
- DEMUX Types
 - 1-to-2 (1 select line)
 - 1-to-4 (2 select lines)
 - 1-to-8 (3 select lines)
 - 1-to-16 (4 select lines)

Demultiplexer
Block Diagram

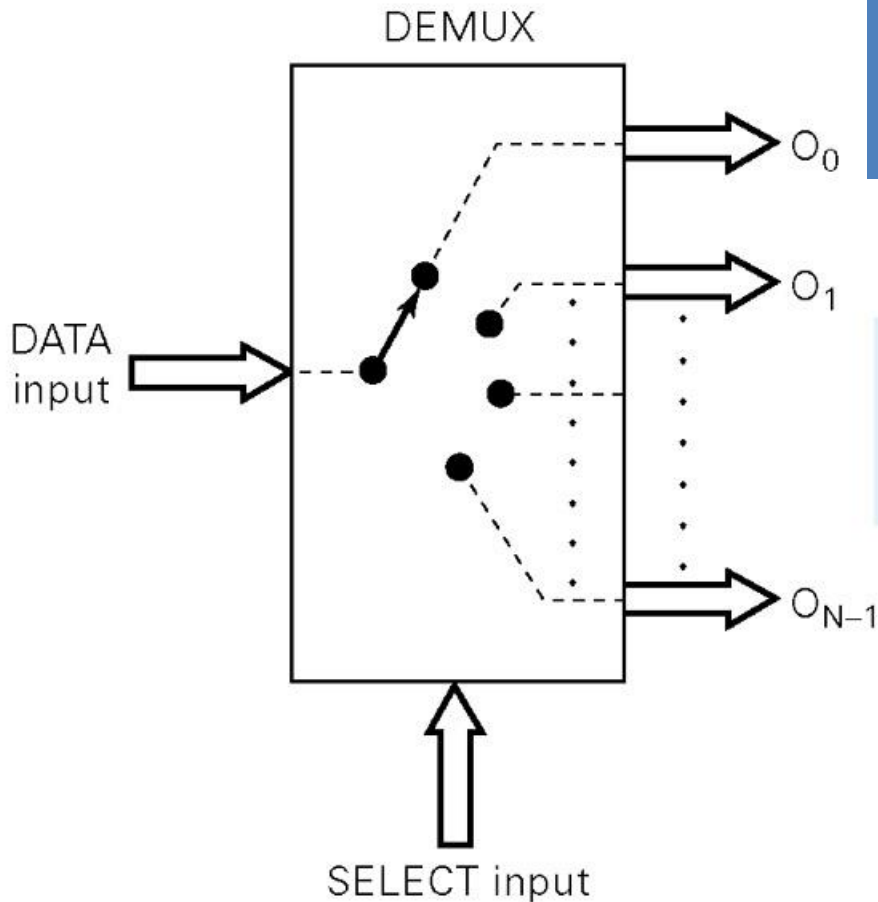


Typical Application of a DEMUX



General demultiplexer

Functional diagram:-

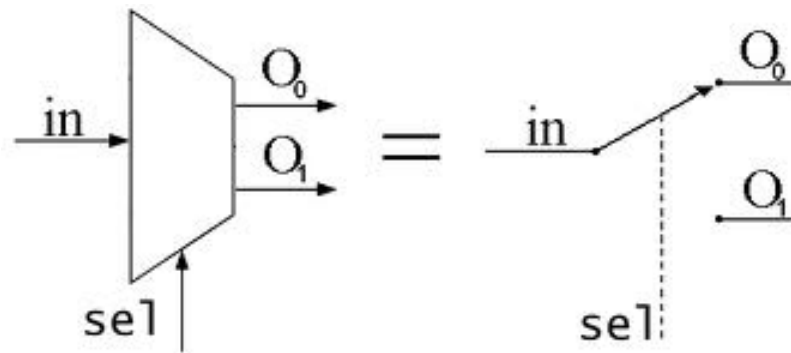


The large arrow indicates one or more lines. The select i/p code determines to which output the DATA input will be transmitted

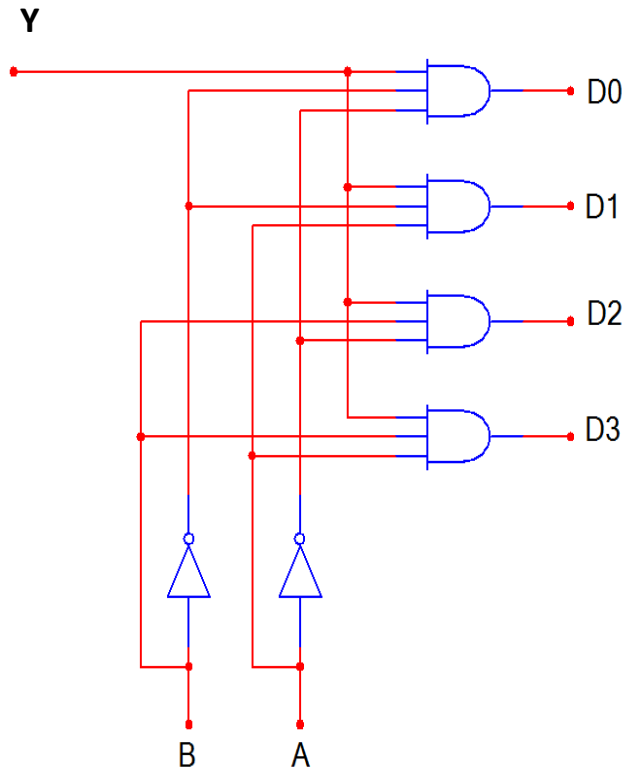
DATA input is transmitted to only one of the outputs as determined by select input code

In other words, the demultiplexer takes one data input source and selectively distributes it to 1 of N output channels just like multiposition switch.

1-to-2 De-Multiplexer



1-to-4 De-Multiplexer (DEMUX)

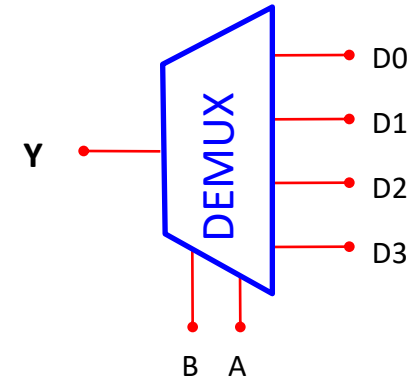


$$D0 = A'.B'.Y$$

$$D1 = A.B'.Y$$

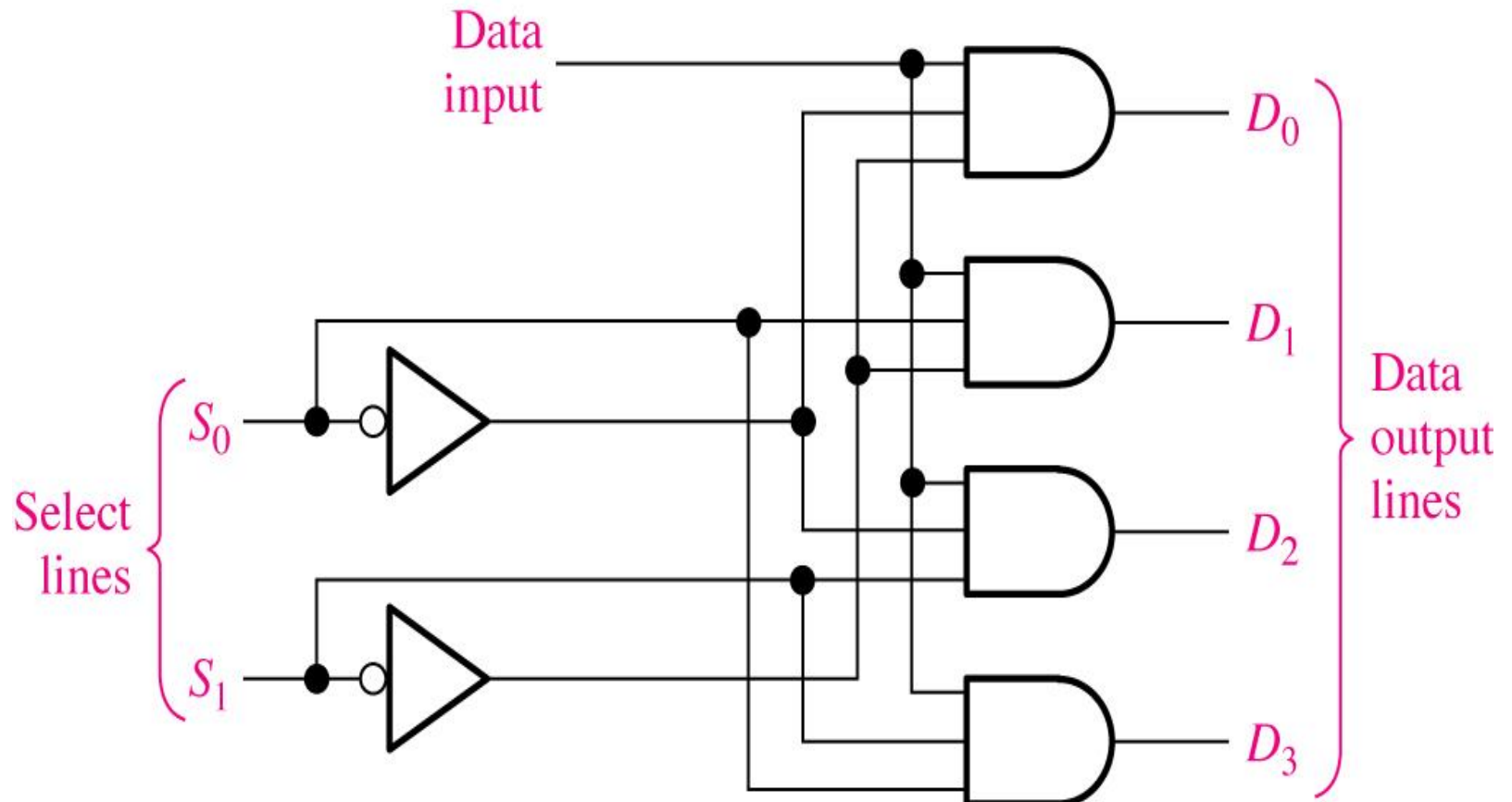
$$D2 = A'.B.Y$$

$$D3 = A.B.Y$$

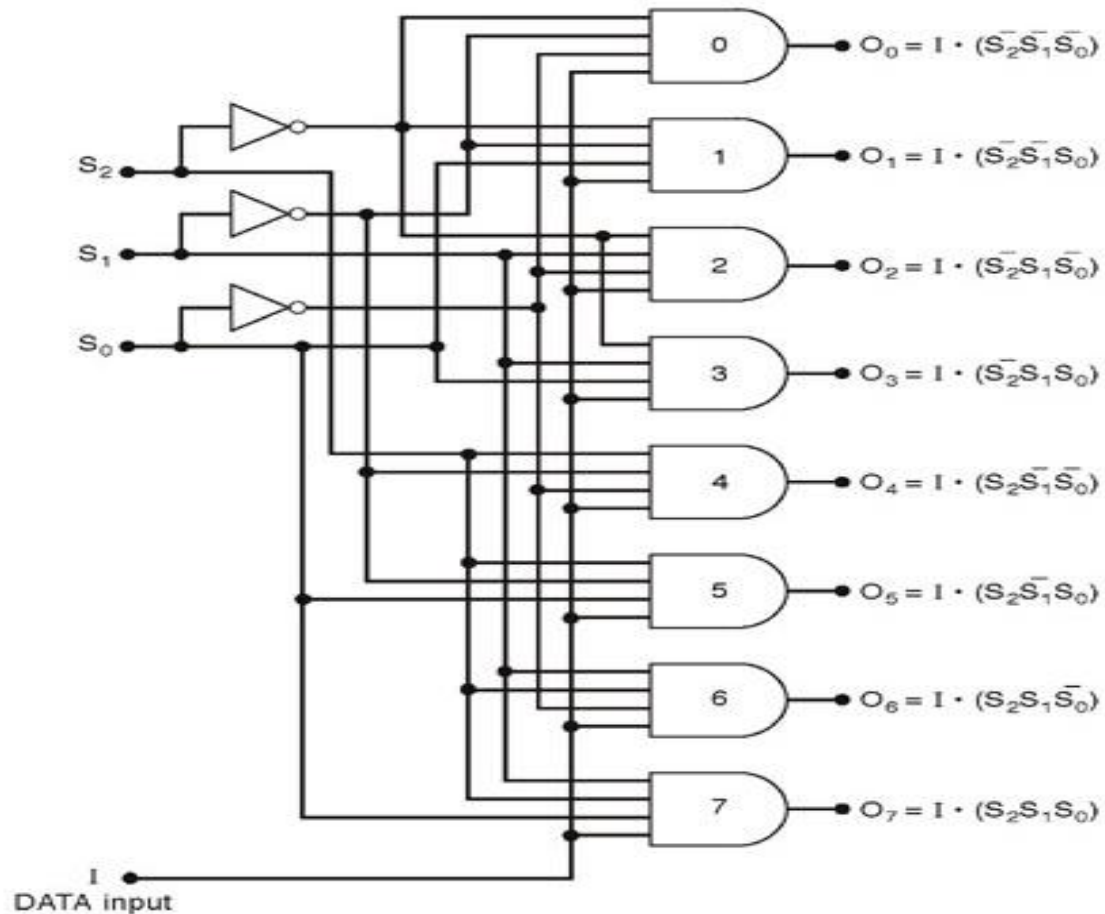


B	A	D0	D1	D2	D3
0	0	Y	0	0	0
0	1	0	Y	0	0
1	0	0	0	Y	0
1	1	0	0	0	Y

A 1-line-to-4-line demultiplexer.



1-line- to-8 line demultiplexer

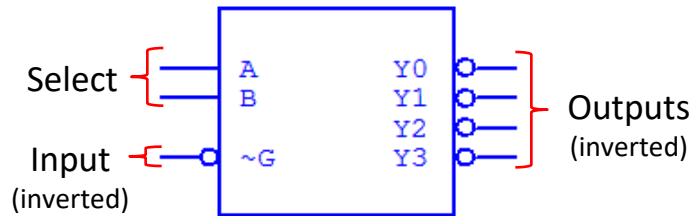


SELECT code			OUTPUTS							
S_2	S_1	S_0	O_7	O_6	O_5	O_4	O_3	O_2	O_1	O_0
0	0	0	0	0	0	0	0	0	0	I
0	0	1	0	0	0	0	0	0	I	0
0	1	0	0	0	0	0	0	I	0	0
0	1	1	0	0	0	0	I	0	0	0
1	0	0	0	0	0	I	0	0	0	0
1	0	1	0	0	I	0	0	0	0	0
1	1	0	0	I	0	0	0	0	0	0
1	1	1	I	0	0	0	0	0	0	0

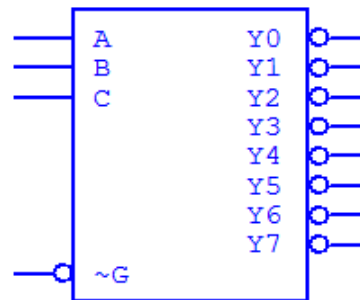
Note: I is the data input

Medium Scale Integration DEMUX

1-to-4 DEMUX



1-to-8 DEMUX



16-to-1 MUX

